These results contribute to a deeper understanding of learning dynamics in neural networks and offer a new tool for anticipating and potentially inducing generalization in machine learning models.

Our experiments on modular arithmetic tasks (x_div_y, x_minus_y, x_plus_y) demonstrated successful grokking, with validation accuracies reaching 100% (Table 1). The permutation task, however, showed limited generalization with a final validation accuracy of 33.93%, highlighting the challenges in applying our approach to more complex scenarios.

The strong correlation between MDL reduction and validation accuracy improvement, as shown in Figure 3, supports the hypothesis that compression of internal representations plays a crucial role in the grokking process. Figure 2 further illustrates the clear relationship between MDL transition points and grokking points across different tasks.

While our results are promising, limitations and areas for future work include:

1. Extending the study to more complex problems and larger-scale neural networks. 2. Exploring the application of our MDL estimation technique to diverse datasets in natural language processing and computer vision. 3. Investigating the relationship between MDL and other generalization metrics. 4. Developing training algorithms that explicitly optimize for MDL reduction alongside traditional loss functions. 5. Examining the interplay between MDL, grokking, and other phenomena such as double descent. 6. Incorporating other compression-based metrics and information-theoretic measures for a more nuanced understanding of generalization in neural networks.

In conclusion, our work provides a novel information-theoretic perspective on the grokking phenomenon, opening new avenues for understanding and improving generalization in deep learning. As the field continues to evolve, we believe that information-theoretic approaches like the one presented in this paper will play an increasingly important role in unraveling the mysteries of neural network learning and generalization.

8 Related Work

CAUTION: THIS PAPER WAS AUTONOMOUSLY GENERATED BY THE AI SCIENTIST

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Review

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"Summary": "This paper investigates the phenomenon of grokking in neural
networks through the lens of Minimal Description Length (MDL), offering an
information-theoretic perspective on sudden generalization. The authors
propose a method to estimate and track MDL during training using weight
pruning techniques. Experiments on modular arithmetic and permutation tasks
reveal a strong connection between MDL transitions and grokking points,
with varying dynamics across different tasks.",
"Strengths": [
    "The paper addresses a significant and poorly understood phenomenon in
neural networks, grokking.",
    "The use of Minimal Description Length (MDL) to analyze grokking is
novel and provides valuable insights.",
    "The experimental results on modular arithmetic tasks are strong,
showing clear connections between MDL reduction and generalization.",
    "The paper introduces new visualization techniques for understanding
the relationship between MDL and grokking."
],
"Weaknesses": [
    "The description of the weight pruning technique and how MDL is
estimated lacks clarity and detail.",
    "The poor performance on permutation tasks raises questions about the
generalizability of the findings.",
    "The theoretical grounding of the connection between MDL and grokking
could be strengthened.",
    "The experimental setup is not comprehensive enough, with limited
datasets and tasks.",
    "The significance of the results for practical applications in neural
network training and model design is not well-articulated."
],
"Originality": 3,
"Quality": 2,
"Clarity": 2,
"Significance": 3,
"Questions": [
    "Can the authors provide a more detailed description of the weight
pruning technique and how MDL is estimated?",
    "What are the potential reasons for the poor performance on permutation
tasks, and how might the approach be improved?",
    "Can the authors provide more theoretical grounding for the connection
between MDL and grokking?",
    "How is the weight pruning technique implemented for MDL estimation,
and why was the specific threshold chosen?",
    "Can the authors extend their experiments to more complex and diverse
tasks to test the generalizability of their findings?",
    "What are the practical implications of these findings for neural
network training and model design?"
],
"Limitations": [
    "The paper needs to address the clarity of the description of methods,
particularly weight pruning and MDL estimation.",
    "The generalizability of the findings beyond modular arithmetic tasks
is questionable based on the results for permutation tasks.",
    "The potential negative societal impacts of this work are not
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discussed, although the focus on theoretical and empirical analysis may
have minimal direct societal consequences."
],
"Ethical Concerns": false,
"Soundness": 2,
"Presentation": 2,
"Contribution": 2,
"Overall": 3,
"Confidence": 4,
"Decision": "Reject"
```